



The quant platform that tells you the truth.

Detailed Feature Guide

All 11 core features of the platform, explained - what each one does, how it works under the hood, and why it matters to a trader.

Prepared for review. Runs out of the box in demo mode - synthetic data, no API keys required.

About AlphaForge

AlphaForge is a quantitative trading research and decision platform. It takes the tools professional quant desks use - signal generation, rigorous backtesting, risk analytics, portfolio construction - and packages them for ordinary traders, with two things almost no retail tool offers: an engine that refuses to lie about a strategy's real edge, and an AI layer that lets a non-expert drive the whole thing in plain English.

The eleven features below make up the platform. The first four are its signature capabilities; the rest are the professional-grade analytics underneath.

- 01 Honesty Engine** - the moat - is the edge real, or luck?
- 02 AI Strategy Lab** - plain English in, an honest verdict out
- 03 Strategy Backtester** - rigorous, lookahead-free backtesting
- 04 Paper Trading** - forward-test on unseen data before real money
- 05 Dashboard / Command Center** - portfolio risk & return at a glance
- 06 Signals** - what the rules say to do right now
- 07 Risk Analytics** - know your downside before it happens
- 08 Portfolio Optimization** - build a portfolio, not just single bets
- 09 Factor Analytics** - understand what actually drives returns
- 10 Regime Detection** - adapt to bull, bear and choppy markets
- 11 Prediction / Forecasting** - statistical & ML forecasts, honestly caveated

The 11 features in detail

01

Honesty Engine

THE MOAT | Answers the one question every backtest hides: is this edge real, or an illusion of luck and overfitting?

- What it does** Grades a strategy and returns a blunt verdict - PROBABLY REAL, LIKELY OVERFIT or INCONCLUSIVE - plus two plain answers: would this have blown up your account, and does it beat simply buying and holding after costs?
- How it works** It runs every backtest through the academic overfitting toolkit: the Deflated Sharpe Ratio (which discounts your Sharpe for how many strategy variations were tried and for non-normal returns) and the Probability of Backtest Overfitting via combinatorial cross-validation (how often the best in-sample setup fails out-of-sample). It also runs a ruin check on the worst drawdown.
- Why it matters** Almost no retail tool does this. It is the single feature that turns an impressive-looking demo into a trustworthy product - it stops you fooling yourself before you risk money.

02

AI Strategy Lab

ACCESSIBILITY | Describe a strategy in ordinary English and get a runnable, tested, honestly-graded result back - no coding.

- What it does** You type an idea like 'buy when RSI drops below 30, sell at 70'. AlphaForge interprets it, tests it, shows the Honesty Verdict, and returns a plain-language risk audit with sources.
- How it works** A built-in parser maps your words to one of five tested strategies and extracts the parameters (with an optional AI model for messier descriptions, always validated back onto tested rules). It runs a real backtest, grades it with the Honesty Engine, and an AI risk auditor turns the numbers into plain warnings - each grounded in cited quant-finance research, not hallucinated.
- Why it matters** This is the accessibility unlock: it lets someone with no quant background drive the entire platform and, crucially, be told the truth about their idea in language they understand.

03

Strategy Backtester

CORE | Full-control backtesting that is correct by construction - so the numbers you see are numbers you can trust.

- What it does** Replays a strategy over history with realistic costs, then validates it three ways: walk-forward testing, Monte Carlo simulation, and the Honesty Verdict. Includes a regime-aware mode that auto-switches strategy by market condition.
- How it works** The engine avoids lookahead bias by design (positions act on the next bar), and models real India (STT, stamp duty, SEBI, GST, slippage) and US cost structures. Walk-forward reports an efficiency ratio (out-of-sample vs in-sample Sharpe) with an overfit warning; Monte Carlo runs 1,000 paths to separate luck from skill (probability of profit, risk of ruin).
- Why it matters** It closes every common way a backtest lies - peeking at the future, ignoring costs, and over-fitting - so a strategy that looks good here has earned it.

04

Paper Trading

FORWARD PROOF | *Forward-test a strategy on data it has never seen, with simulated money, before any real capital.*

- What it does** Deploy a strategy to a paper account and track it forward from the deploy date, watching its equity race against buy-and-hold in real time.
- How it works** It steps bar by bar from the deploy date, acting only on the previous bar's signal so there is no lookahead - every result is genuinely out-of-sample. It runs on a built-in simulator by default (no keys, no real money) and can route to a broker's paper account when configured, which it restricts to paper endpoints for safety.
- Why it matters** A backtest can be fooled; forward performance on unseen data is the closest thing to proof. This is the last honest check between 'the backtest looked great' and 'it actually works'.

05

Dashboard / Command Center

OVERVIEW | *A single screen showing the risk and return of your whole watchlist.*

- What it does** Presents portfolio KPIs (Sharpe, Sortino, CAGR, max drawdown, VaR, CVaR, beta, win rate), an equity curve with rolling Sharpe, current market regime with a bear-probability gauge, the return distribution with VaR/CVaR marked, active signals, an asset correlation heatmap, and a market snapshot table.
- How it works** Every panel is computed live from the shared data engine using the same metrics library that powers the rest of the app, so the numbers are consistent everywhere.
- Why it matters** It gives an immediate, honest read on the health of your book - before you drill into any single strategy.

06

Signals

TIMING | *Shows what your indicator rules say to do right now, across all your tickers.*

- What it does** Scans each ticker for RSI, MACD and Bollinger-band signals, combines them into a single buy/hold/sell stance, and can explain the reasoning in plain language.
- How it works** Signals are generated by the same canonical indicator functions the backtester uses, so what you see live matches what gets tested - no hidden discrepancy between screen and simulation.
- Why it matters** It turns a wall of indicators into one clear, current answer per ticker, so you act on rules rather than emotion.

07

Risk Analytics

DOWNSIDE | Quantifies how much you could lose - using methods that respect fat tails, not just the bell curve.

- What it does** Computes Value at Risk and Conditional VaR several ways (historical, parametric, Student-t and GARCH), tests whether the VaR model is actually accurate, and analyses drawdowns and portfolio-level risk.
- How it works** The Student-t and GARCH methods capture the fat tails and volatility clustering that real markets show and the normal distribution misses; a Kupiec proportion-of-failures test then checks, statistically, whether the risk model has been under-estimating danger.
- Why it matters** Most blow-ups come from ignored tail risk. This makes the downside explicit and honest before it arrives.

08

Portfolio Optimization

CONSTRUCTION | Helps you build a diversified portfolio instead of betting on single positions.

- What it does** Generates an efficient frontier of risk-versus-return via Monte Carlo, computes risk-parity weights, and reports portfolio-level statistics.
- How it works** It uses the covariance structure across assets, so it accounts for how holdings move together - risk parity in particular sizes positions by risk contribution rather than by capital.
- Why it matters** Diversification is one of the few genuinely free lunches in markets; this makes it concrete and measurable.

09

Factor Analytics

DRIVERS | Reveals what actually drives returns - momentum, value, size, quality, low volatility.

- What it does** Builds a cross-sectional factor matrix and measures each factor's predictive power through the Information Coefficient, its reliability (ICIR), quintile-sorted returns and how the signal decays over time.
- How it works** The Information Coefficient is the rank correlation between a factor's score and forward returns; tracking it (and its stability) shows whether a factor has real, persistent predictive value or is just noise.
- Why it matters** It moves you from 'this stock feels strong' to 'this measurable characteristic has historically predicted returns'.

10

Regime Detection

CONTEXT | *Identifies whether the market is trending up, trending down, or choppy - and how strategies behave in each.*

- What it does** Classifies the market into regimes using both a Hidden Markov Model and a fast rule-based bull/sideways/bear classifier, and reports regime-conditional performance.
- How it works** Strategies are regime-dependent: trend-following thrives in trends and bleeds in chop, while mean-reversion is the opposite. Detecting the current regime - and knowing which strategy suits it - explains much of real-world performance.
- Why it matters** It stops you judging a strategy on the wrong conditions, and feeds the backtester's regime-aware auto-switch mode.

11

Prediction / Forecasting

FORECASTS | *Statistical and machine-learning forecasts - offered with honest caveats about their limits.*

- What it does** Provides ARIMA and GARCH statistical forecasts and machine-learning models (such as LSTM), with clear framing that short-series price forecasting overfits easily.
- How it works** Each model is wrapped by the same clean data pipeline and evaluated consistently; the platform deliberately does not add ever more forecasters, because quantity here hurts rather than helps.
- Why it matters** Forecasts can inform a view, but AlphaForge treats them as one input among many - never as a crystal ball - consistent with its honesty-first promise.

Underneath & alongside

Supporting the eleven features are the foundations they all share:

- **Data engine** - clean OHLCV loading with timezone/dedup handling and self-contained demo-mode synthetic data.
- **Metrics & indicators libraries** - the shared, tested maths every feature is built on.
- **Alerts** - price and indicator alerts with email notifications and an alert log.
- **Graphs & Auditing** - deeper research charts and data-quality checks.
- **REST API** - around 32 endpoints wrapping the core, so the whole platform can be driven programmatically.

One promise ties all eleven together.

Every feature is built on the same principle: show the truth, not a flattering picture. The analytics act as the brain; honesty is the conscience that keeps them useful.

Disclaimer: AlphaForge is an educational research tool, not financial advice. It grades strategies for honesty - it does not predict the future. Demo mode uses synthetic data and no real money is ever involved.